

Chemistry Ch 3.2: Subatomic Particles

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Particle	Charge	Mass
Proton	+1	~1 amu (1.6726×10^{-27} kg)
Neutron	0	~1 amu (1.6749×10^{-27} kg)
Electron	-1	~ 1/1836 amu (9.11×10^{-31} kg)

Mass ka asal nuqta

Proton aur neutron ka mass taqreeban **barabar** hai. Lekin electron ka mass sirf **1/1836** hissa hai. Isi liye atom ka **poora mass nucleus** mein hai. Electrons ka mass ginti mein hi nahin aata.

Nucleus aur nuclear force

Nucleus mein **protons aur neutrons** hote hain, aur dono ko mila kar **nucleons** kehte hain. Ab sawal: saaray protons positive hain, to ek doosre ko dhakelna chahiye. Phir nucleus tootta kyun nahin?

Jawab: **nuclear force** (strong force), jo protons aur neutrons ko aapas mein jakar rakhti hai.

Electric field mein teenon ka raweyya

Protons negative plate ki taraf murte hain (kyunke wo positive hain).

Electrons positive plate ki taraf murte hain.

Neutrons bilkul seedhe chale jate hain, kyunke un par charge hai hi nahin.

Aur ek baat: neutral atom mein **protons ki tadaad = electrons ki tadaad**, isi liye atom par koi charge nahin.

Quiz

1. Make a table of the three subatomic particles with their charge and mass. Then explain why almost all the mass of an atom is in the nucleus.
2. How does each of the three behave in an electric field, and why? What holds the nucleus together despite the protons repelling each other?